

Product Requirements Document (PRD)

1. Document Information

Product/Feature Name: Smart Scheduler
Author: Melodi Parton
Date Created: 2025-09-18
Last Updated: 2025-09-18
Version: 0.1 (Draft)

2. Overview

Summary: Smart Scheduler will be a workforce management tool designed for dispatch centers and shift-based operations. It will provide a centralized system for managing staff schedules, tracking overtime, and monitoring certification compliance. By replacing fragmented spreadsheets and manual processes, Smart Scheduler will reduce errors, save administrative time, and ensure operational compliance.

Problem Statement: Current scheduling relies on spreadsheets and paper forms, which are prone to problems like, uncovered shifts, or hours worked beyond regular shifts, and overlooked certification expirations. This leads to increased overtime costs, compliance risks, and wasted time resolving errors.

Goals & Objectives:

- Automate shift scheduling, by allowing employees to show that they are available to work on their scheduled days off, helped to avoid conflicts and coverage gaps.

- Tracking overtime hours in real time to prevent burnout and budget overruns.

- Monitoring certifications and alerting managers when renewals are required.

- Providing a single interface, accessible via the web or a mobile app, for employees to show availability, request PTO or shift swaps, and managers to approve them in a timely manner.

Non-Goals:

- Full payroll processing or timecard integration.

- Predictive AI-driven workforce analytics (future iteration).

- Role-based scheduling permissions (beyond basic manager/staff separation).

3. Context & Background

Business Context: Dispatch centers face compliance and staffing challenges. With increasing safety and regulatory requirements, missing a certification or failing to provide adequate coverage poses financial and legal risks.

Market/Customer Insights:

- Users: Supervisors, and employees.

- Last-minute shift coverage can lead to excess overtime. Having a system that shows ahead of time if an employee is available during their scheduled time off can help resolve

the issue much quicker.

Competitive/Benchmark References:

Competitors include Kronos Workforce Ready and WhenToWork, but these are costly and not tailored to small-to-mid dispatch operations. Smart Scheduler offers a lightweight, purpose-built alternative.

4. Scope

In Scope:

- Create, update, swap and delete shifts.

- Track and display overtime hours.

- Manage certification expiration dates.

- Generate compliance alerts (basic).

Out of Scope:

- Payroll system integration.

- SMS or phone-based scheduling notifications.

- AI-based scheduling optimization.

5. User Stories & Use Cases

Primary User Personas:

- Dispatch Manager: Oversees scheduling, compliance, and overtime.

- Staff Member: Views shifts, requests changes, monitors personal certifications.

User Stories:

- As a manager, I want to create and assign shifts so that all coverage is filled without conflicts.

- As a manager, I want to see overtime hours per employee so that I can prevent burnout.

- As a manager, I want to receive alerts when certifications are expiring so I can keep my team compliant.

- As a staff member, I want to view my schedule so I know when I am working.

- As a staff member, I want to show when I am available for extra shifts.

Use Case Scenarios:

- Happy Path: A manager creates shifts, employees see them in their dashboard, employees can see empty shifts and show if they are available or not, and alerts appear when a certification is nearing expiration.

- Edge Case: If two employees are accidentally scheduled for the same shift; the system warns and prevents the conflict.

- Edge Case: If two employees show they are available for an open shift, the manager can schedule the one with the least hours to prevent excess overtime for one employee.

6. Functional Requirements

FR-001: System must allow managers to create, edit, and delete shifts.

FR-002: System must track and display overtime hours per employee.

FR-003: System must record certification dates and generate alerts before expiration.
FR-004: Staff must be able to view their assigned shifts in real time.
FR-005: Staff must be able to show their availability for unscheduled shifts.

7. Non-Functional Requirements

Performance: Pages should load within 2 seconds under normal load.
Scalability: Support up to 200 concurrent users in MVP phase.
Accessibility: Compliant with WCAG 2.2 AA (keyboard navigation, color contrast).
Security/Compliance:
 User authentication via Django's auth system.
 Data protection for employee personal info.
Reliability/Availability:
 99% uptime target for MVP.

8. Dependencies

Internal: Django framework, SQLite/PostgreSQL database.
External: GitHub repository for code versioning.
Future: Potential integration with certification authority APIs.

9. Risks & Assumptions

Risks:
 Staff resistance to replacing paper workflows.
 Certification data may not always be up to date.
 Scope creep if payroll or advanced features are requested early.
Assumptions:
 All users have access to internet and modern browsers.
 Dispatch managers will be responsible for data entry.

10. Acceptance Criteria

AC-001: Manager can create and publish a shift that displays correctly in staff view.
AC-002: Employees can show availability for shifts that display open.
AC-003: Overtime report shows correct totals when employees exceed 40 hours.
AC-004: Certification expiration alerts appear 30 days before the due date.

11. Success Metrics

80% reduction in scheduling conflicts.
50% less time spent on schedule updates.
100% compliance with certification renewals tracked in the system.
User satisfaction $\geq 85\%$ in surveys after pilot rollout.

12. Rollout & Release Plan

Phasing:

MVP: Basic scheduling + certification tracking.

Phase 2: Overtime dashboards.

Phase 3: Notifications and reporting features.

Release Channels: Pilot with single dispatch center, then scale.

Training/Documentation: User guide PDF and short training video.

13. Open Questions

Do we need exportable reports for compliance audits?

Should the overtime threshold be configurable per agency?

14. References

Layman, Matt. Understand Django (Ch. 3–4: Views and Templates).

Jennex, M. Why Computers Are Vulnerable (cybersecurity reference).

Internal Pitch Memo: docs/Pitch.pdf.